

Products close to a perfect power

Problèmes ouverts — A. Dujella · 2026

Problem 1.6 (Dujella, 2026). Prove unconditionally that the size of a set of positive integers with the property that the product of any two of its distinct elements is 1 less than a perfect power is bounded by an absolute constant.

Remark. Luca (Glas. Mat. Ser. III **40** (2005), 13–20) proved this result under the assumption that the abc-conjecture is valid.